

Cohort 2, Team #3

# Radio Frequency (RF) – Induced Voltage Detector for Active Implantable Medical Device MR safety analysis

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## **Abstract**

This report presents the application of the ADDIE (Analysis, Design, Development, Implementation, Evaluation) framework to the design of a portable RF-induced voltage detector for Active Implantable Medical Device (AIMD) testing in MRI environments. During this semester, the project progressed through the Analysis, Design and Development phases. The system is intended to measure induced voltages on AIMD leads under RF exposure conditions, providing real-time display and data logging capabilities. Key design considerations include RF noise mitigation, signal integrity, usability, and compliance with MRI-compatible constraints. The report summarizes system requirements, architecture, trade studies, and planned next steps toward prototype development and validation.

## **I. Introduction**

Active Implantable Medical Devices (AIMDs) are widely used in modern healthcare, yet their interaction with MRI environments presents significant safety concerns. Exposure to strong RF fields can induce voltages on device leads/IPG (implantable pulse generator)s, potentially resulting in heating, unintended stimulation, or device malfunction. Reliable measurement of these induced voltages is essential for safety validation.

This project aims to design a portable detector capable of accurately measuring RF-induced voltages on AIMD lead/IPGs in a controlled testing environment. The ADDIE framework is used to structure the design process. This report documents the completion of the Analysis and Design phases and outlines the transition to Development.

## **II. Analysis Phase**

### **A. Problem Definition**

MRI environments expose AIMDs to strong RF fields, leading to several hazards including electrode heating, unintended stimulation, and device malfunction. Existing testing methods are often expensive, complex, or inaccessible, creating a need for a practical and cost-effective measurement solution.

### **B. Stakeholders and Users**

The primary stakeholder for this project is the FDA, with end users consisting of medical researchers and laboratory technicians. These users are expected to have medical expertise but limited background in electrical engineering, necessitating a system that is simple to operate and provides clear, interpretable outputs.

### **C. System Requirements**

The detector must satisfy the following requirements:

- Voltage measurement range: 0.5 V to 14 V
- Measurement precision:  $\pm 0.1$  V
- Sampling rate:  $\geq 20$  kS/s
- Operation in MRI RF environments (64 MHz and 128 MHz)
- Real-time display and data logging
- Battery-powered operation (up to 8 hours)

### **D. Constraints**

The design is subject to several constraints:

- MRI-compatible materials (non-ferrous)
- Cable lengths below 30 cm to prevent antenna effects - can be above 30 cm but common-mode-currents on the cable need to be suppressed.
- Waterproof enclosure for gel-based testing environments
- Separation of analog and digital power domains in the circuit board.
- Reliable operation under strong RF environment

### **E. Environment and Resources**

Due to limited access to MRI machines, testing will be performed using EMF generators and tissue-simulating phantoms. AIMD samples are provided for validation purposes.

## **III. Design Phase**

### **A. System Architecture**

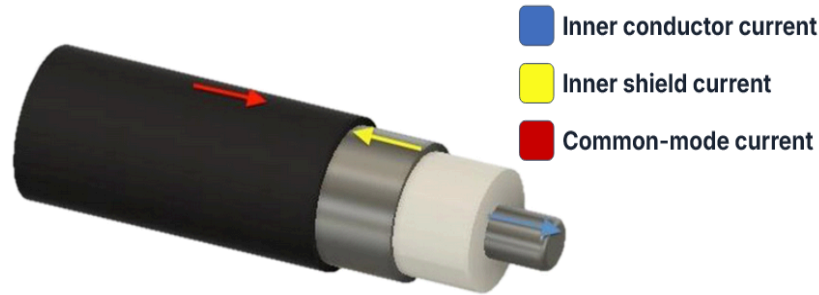
The system follows a structured signal-processing pipeline:

1. Voltage sensing from AIMD leads
2. Transmission via twisted pair and coaxial cable
3. RF suppression using a balun
4. Signal conditioning (amplification, rectification, filtering)
5. Analog-to-digital conversion
6. Digital processing and output

The architecture ensures accurate measurement while minimizing RF-induced distortions.

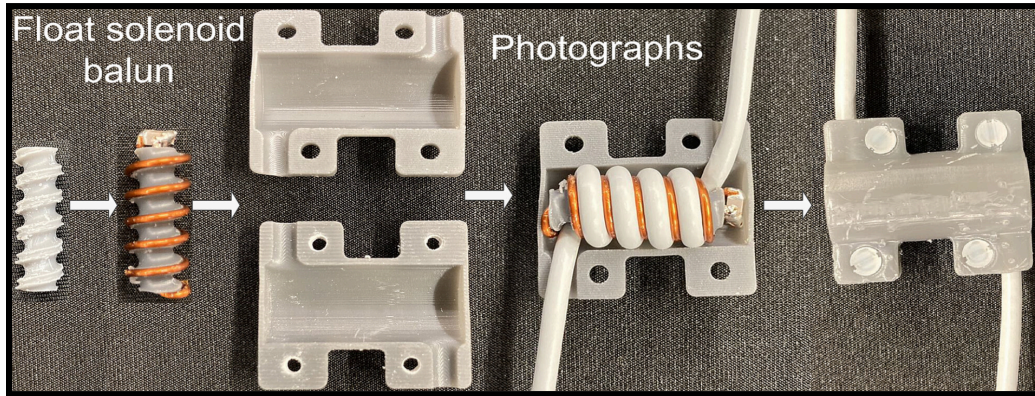
## B. Signal Integrity and RF Mitigation

Because the detector operates inside the strong RF environment, the measurement cable is not electrically passive. The B1+ field at 64 MHz (1.5T) and 128 MHz (3T) induces common-mode currents on the outer surface of the coaxial shield as seen in Figure 1, effectively turning the cable into an unintended antenna. These currents propagate independently of the desired differential signal inside the cable and can lead to measurement distortion, unintended RF coupling, and localized heating. As a result, the measured voltage at the AIMD terminals no longer reflects the true induced voltage, but rather a combination of the device response and cable-induced artifacts.



*Figure 1. Coax diagram of Common-Mode Current.*

To preserve measurement integrity, the system must include a mechanism that suppresses these shield currents without affecting the internal signal path. Therefore, a cable trap is introduced as a critical component of the signal chain. The trap operates by presenting a high impedance to common-mode currents on the cable shield, while remaining effectively transparent to the desired differential signal.



*Figure 2. Floating solenoid balun architecture chosen for VI.*

As illustrated in Figure 2, the selected implementation is a floating solenoid balun (FSB), which inductively couples to the shield current and reflects a high impedance at resonance, thereby blocking RF current propagation along the cable.

### C. Signal Conditioning and Peak Detection

In order for the ADC to properly measure the induced voltages, the low frequency amplitude fluctuations must be extracted from the high frequency input signal (64 or 128 MHz). Below are two possible circuit designs for this application.

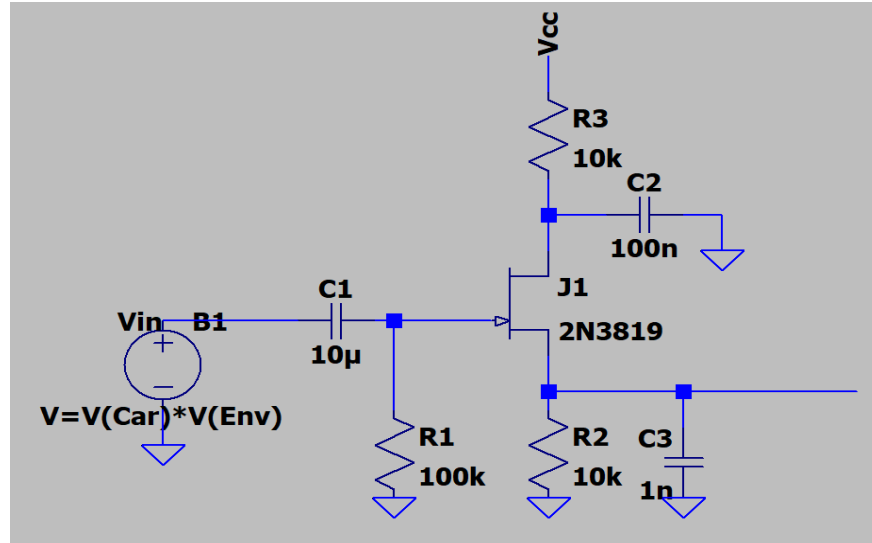


Figure 3. 2N3819 JFET based peak detector

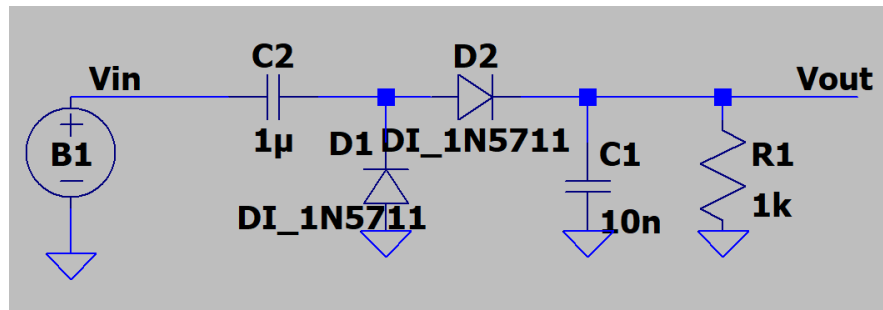


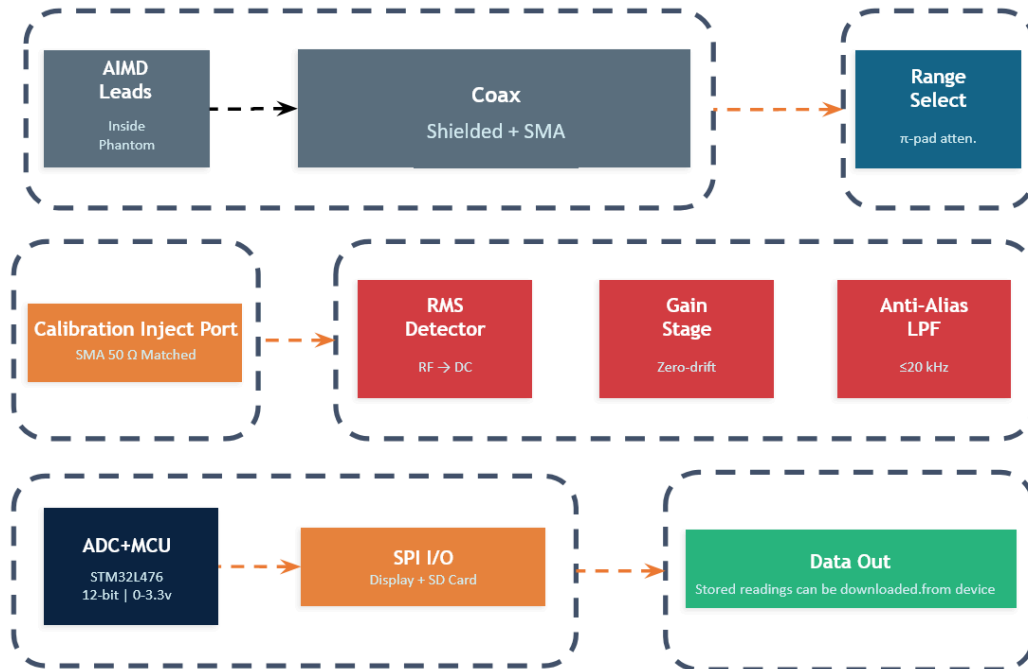
Figure 4. 1N5711 Diode based peak detector

### D. Hardware Design

The hardware system consists of:

- Analog front-end (differential amplifier, rectifier, filter)
- STM32 microcontroller for data acquisition and processing
- LCD module for real-time display
- SD card module for data storage
- Battery power system

Careful separation of analog and digital subsystems is implemented to reduce noise coupling.



*Figure 5. Hardware Architecture Block Diagram.*

## E. Software Design

The embedded software is structured as a finite state machine (FSM) to manage:

- Signal acquisition
- Data processing
- Display updates
- Data logging
- Error detection and reporting

This structure ensures reliable and predictable system behavior.

# Software FSM

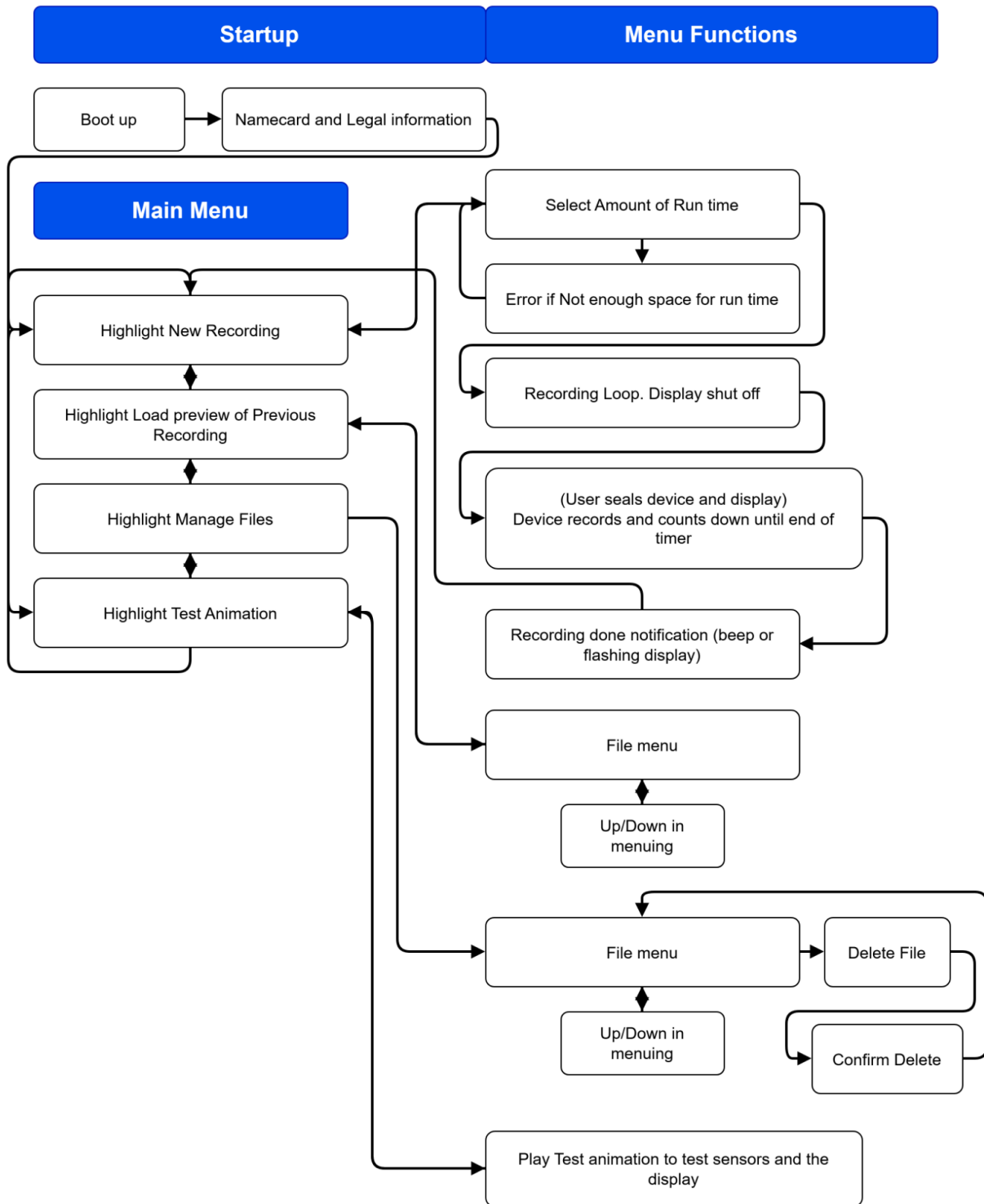


Figure 6. Software FSM.

## F. Engineering Specifications

The system is designed to meet the following specifications:

- Voltage range: 0.5 V to 14 V
- Precision:  $\pm 0.1$  V
- Sampling rate:  $\geq 20$  kS/s
- Runtime: up to 8 hours
- Memory buffering: minimum 2 seconds
- SPI communication with parity

## G. Trade Studies

Trade studies were conducted to evaluate design alternatives for:

- Cable trap implementation

This cable trap approach was selected based on the trade study results shown in Fig 5, the study prioritized common-mode suppression, measurement integrity, and buildability. The FSB design preserves the physical integrity of the coaxial cable, allows repositioning during testing, and enables tuning to the target MRI frequencies, making it the most practical and effective first-stage solution for the detector system, closely followed by the Dual-Tuned FSB which will be the focus of the second prototype.

|                         | CMRR<br>30% |                     | REVERSIBLE<br>20% |                      | FLEXIBILITY<br>10% |            |      |
|-------------------------|-------------|---------------------|-------------------|----------------------|--------------------|------------|------|
| Floating Solenoid Balun | 9           | 8                   | 10                | 6                    | 5                  | 9          | 8.20 |
| Dual-Tuned FSB          | 8           | 6                   | 7                 | 10                   | 4                  | 8          | 7.35 |
| Printed Bazooka Balun   | 7           | 6                   | 3                 | 6                    | 8                  | 9          | 6.15 |
| Caterpillar Trap        | 6           | 4                   | 2                 | 6                    | 10                 | 8          | 5.65 |
|                         |             | BUILDABILITY<br>20% |                   | FREQ COVERAGE<br>15% |                    | LOSS<br>5% |      |

Figure 7. Cable-Trap Trade Study Matrix.

The validation plan shown in Fig 6 focuses on confirming that the cable trap effectively suppresses common-mode currents without degrading the measurement signal. A floating solenoid balun (FSB) will be built and tuned to 64 MHz and 128 MHz, then evaluated using VNA or current-probe measurements to compare cable behavior with and without the trap. Additional tests will assess placement sensitivity and verify minimal impact on the desired

signal. Success is defined by clear resonance at target frequencies, measurable attenuation of common-mode currents, and stable performance suitable for system integration.

|                                                         |                                                                       |                                                         |                                                              |                                                   |
|---------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------|
| STEP 01                                                 | STEP 02                                                               | STEP 03                                                 | STEP 04                                                      | STEP 05                                           |
| <b>Build a single FSB</b>                               | <b>Tune resonance</b>                                                 | <b>Measure blocking</b>                                 | <b>Scan placement</b>                                        | <b>Repeat at 2nd freq</b>                         |
| Non-magnetic parts,<br>removable mechanical<br>housing. | Adjust coil / capacitor<br>values toward the<br>selected target freq. | VNA / current-probe:<br>cable with vs. without<br>trap. | Slide trap along cable;<br>record changes in<br>suppression. | Decide whether<br>dual-tuned FSB is<br>justified. |

Figure 8. Benchtop Validation Plan.

Trade studies were conducted to evaluate design alternatives for:

- Battery Selection and Power Circuit

*Score scale: 10 = best fit, 5 = acceptable, 1 = poor fit. Totals are weighted scores out of 10.*

#### Trade Study 1: Battery Source Selection

| Criteria                          | Weight      | Alkaline C-cell | NiMH C-cell | Li-ion Pack |
|-----------------------------------|-------------|-----------------|-------------|-------------|
| <b>Runtime / Capacity</b>         | 25%         | 7               | 8           | 10          |
| <b>Safety / Management</b>        | 15%         | 9               | 8           | 7           |
| <b>Recharge / Reuse</b>           | 15%         | 1               | 10          | 10          |
| <b>Regulation Noise</b>           | 15%         | 9               | 8           | 7           |
| <b>MRI Material Compatibility</b> | 15%         | 5               | 6           | 7           |
| <b>Packaging / Size</b>           | 15%         | 4               | 5           | 10          |
| <b>TOTAL</b>                      | <b>100%</b> | <b>5.95</b>     | <b>7.70</b> | <b>8.55</b> |

*Recommended: The lithium-ion battery pack is the recommended battery source because it provides the best balance of runtime, rechargeability, compact size, and system integration. Since the detector is used in an MRI-adjacent environment, the selected Li-ion pack, holder, terminals, wiring, and protection board must be checked for magnetic materials before testing. The pack should also include proper protection circuitry, and the power rails should use filtering, RF chokes, and bypass capacitors to reduce RF noise coupling into the DC lines. If magnetic material cannot be verified as safe, the battery should be kept outside the high-field region during early tests.*

#### Trade Study 2: Power Regulation and Rail Architecture

| Criteria                 | Weight      | Shared Rail | Separate Filtered Rails | Separate Batteries |
|--------------------------|-------------|-------------|-------------------------|--------------------|
| Analog Noise Isolation   | 25%         | 3           | 9                       | 10                 |
| RF Filtering on DC Lines | 20%         | 3           | 10                      | 8                  |
| System Simplicity        | 15%         | 10          | 7                       | 4                  |
| Voltage Flexibility      | 15%         | 4           | 9                       | 10                 |
| Integration / Packaging  | 15%         | 10          | 8                       | 3                  |
| Testability              | 10%         | 5           | 8                       | 8                  |
| <b>TOTAL</b>             | <b>100%</b> | <b>5.45</b> | <b>8.75</b>             | <b>7.45</b>        |

*Recommended: one battery with separate filtered analog/digital rails. Add RF chokes in the microhenry range and bypass capacitors on DC rails so DC passes normally while RF noise is blocked or shunted to ground.*

### Trade Study 3: Wiring, Grounding, and RF Mitigation

| Criteria                  | Weight      | Simple Direct Wiring | Controlled Short Wiring | Extensive Shielding |
|---------------------------|-------------|----------------------|-------------------------|---------------------|
| RF Pickup Reduction       | 25%         | 3                    | 8                       | 10                  |
| DC-Line RF Protection     | 20%         | 2                    | 9                       | 9                   |
| Measurement Repeatability | 20%         | 3                    | 9                       | 8                   |
| Build / Debug Ease        | 10%         | 10                   | 7                       | 3                   |
| Return-Current Control    | 15%         | 3                    | 9                       | 8                   |
| Final Prototype Fit       | 10%         | 4                    | 9                       | 7                   |
| <b>TOTAL</b>              | <b>100%</b> | <b>4.00</b>          | <b>8.55</b>             | <b>8.35</b>         |

*Recommended: controlled short wiring with analog/digital separation. Use local decoupling, clean return paths, and selective shielding. A series capacitor can DC-block a signal path, while rail-to-ground capacitors act as RF bypass/decoupling parts.*

- Peak detector design

Table 1 shows a trade study conducted on the two possible peak detector designs shown in Figures 3-4. The table shows a minor difference in the final score for the two options. Due to the simplicity in the designs and the low cost of the circuits, both designs will be constructed for further testing and prototyping.

Table 1. Peak detector trade study

| Criteria             | Weight | 2N3819 JFET | 1N5711 Diode |
|----------------------|--------|-------------|--------------|
| Low Voltage Accuracy | 15%    | 7           | 8            |
| Frequency Range      | 30%    | 10          | 10           |
| Voltage Range        | 20%    | 8           | 7            |
| Power Consumption    | 20%    | 8           | 10           |
| Noise                | 15%    | 8           | 7            |
| Total                | 100%   | 8.45        | 8.65         |

## H. Bill of Materials

A preliminary bill of materials was developed, including RF components, connectors, passive elements, and prototyping hardware. The estimated cost for initial prototyping is approximately \$181.

Table 2. Bill of Materials

| Sponsor Agency                         | Catalog or Parts No.                                | Description                                                                                          | Product Link            | Qty. | Unit Price | Total Price |
|----------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------|------|------------|-------------|
| <i>University of Houston &amp; FDA</i> | 2N3819 PBFREE                                       | JFETs N-CH -25V 10mA BULK 25Vgs 360mW 8pF                                                            | <a href="#">Mouser</a>  | 3    | \$1.89     | \$5.67      |
| <i>University of Houston &amp; FDA</i> | 4526-SP1-50X 50-G                                   | SP1-50X50-G BREADBOARD GENERAL PURPOSE NPTH                                                          | <a href="#">Digikey</a> | 2    | \$4.15     | \$8.30      |
| <i>University of Houston &amp; FDA</i> | uxcell:<br>A19061200ux0 370<br>ASIN:<br>B07V6QNG8 M | uxcell RG316 Coaxial Cable with BNC Male to SMA Male Connectors 50 Ohm 1 Feet 2pcs                   | <a href="#">Amazon</a>  | 1    | \$14.99    | \$14.99     |
| <i>University of Houston &amp; FDA</i> | exgoofit:<br>SMA00104<br>ASIN:<br>B0CNYW6FG F       | SMA Male to SMA Male Cable 6 inch, Pack of 4, RG316 SMA Coaxial Cable Extender for Antenna Extension | <a href="#">Amazon</a>  | 1    | \$6.99     | \$6.99      |
| <i>University of Houston &amp; FDA</i> | Bnafes:<br>8541545872<br>ASIN:<br>B07GXSN7VS        | SMA Female PCB Panel Edge Mount Connector Straight Solder RF Coaxial Adapter 25PCS                   | <a href="#">Amazon</a>  | 1    | \$9.89     | \$9.89      |
| <i>University of Houston &amp; FDA</i> | 511-1N5711                                          | Schottky Diodes & Rectifiers 15mA 70 Volt                                                            | <a href="#">Mouser</a>  | 50   | \$0.06     | \$2.75      |
| <i>University</i>                      | CAT.NO:                                             | 1nF 50VDC Ceramic                                                                                    | <a href="#">Jaycar</a>  | 6    | \$0.45     | \$2.70      |

|                                        |                                                                                |                                                                                                                                                    |                        |   |         |         |
|----------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---|---------|---------|
| <i>of Houston &amp; FDA</i>            | RC5336                                                                         | Capacitors - Pack of 2                                                                                                                             |                        |   |         |         |
| <i>University of Houston &amp; FDA</i> | CAT.NO:<br>RC5348                                                              | 10nF 50VDC Ceramic Capacitors - Pack of 2                                                                                                          | <a href="#">Jaycar</a> | 3 | \$0.55  | \$1.65  |
| <i>University of Houston &amp; FDA</i> | CAT.NO:<br>RC5340                                                              | 2.2nF 50VDC Ceramic Capacitors - Pack of 2                                                                                                         | <a href="#">Jaycar</a> | 3 | \$0.45  | \$1.35  |
| <i>University of Houston &amp; FDA</i> | CAT.NO:<br>RR0593                                                              | 7.5k Ohm 0.5 Watt Metal Film Resistors - Pack of 8                                                                                                 | <a href="#">Jaycar</a> | 1 | \$0.85  | \$0.85  |
| <i>University of Houston &amp; FDA</i> | OPA2376AID R                                                                   | OPA22376                                                                                                                                           | <a href="#">TI</a>     | 4 | \$2.48  | \$9.92  |
| <i>University of Houston &amp; FDA</i> | ASIN:<br>B07T6XWZKN                                                            | DaierTek Waterproof Toggle Switch 12V DC 30A Heavy Duty 2 Pin ON Off SPST with Weatherproof Boot Cap Cover for Auto Car Marine Boat -2 Pack        | <a href="#">Amazon</a> | 1 | \$7.99  | \$7.99  |
| <i>University of Houston &amp; FDA</i> | HiLetgo:<br>3-01-1101-B<br>ASIN:<br>B01HXU1NQY                                 | HiLetgo 10pcs AMS1117-3.3 DC-DC Step-Down Power Module AMS1117-3.3V Buck Module LDO 800MA 3.3V Step Down Regulator                                 | <a href="#">Amazon</a> | 1 | \$6.49  | \$6.49  |
| <i>University of Houston &amp; FDA</i> | Seloky:<br>b0773899-d63<br>3-41e0-97fb-ac<br>cae80afc8a<br>ASIN:<br>B0DBVWQGZL | 10 Pack LM2596 LM2596S Adjustable Voltage Regulator Step Down Variable Volt Regulator Input 3.0-40V Output 1.5-35V DC-DC Buck Converter Electronic | <a href="#">Amazon</a> | 1 | \$12.99 | \$12.99 |

|                                                |                                                |                                                                                                                                                                     |                         |    |        |        |
|------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----|--------|--------|
|                                                |                                                | Voltage Stabilizer<br>Power Supply Module                                                                                                                           |                         |    |        |        |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | 497-6610-1-N<br>D - Cut Tape<br>(CT)           | DIODE SCHOTTKY<br>40V 1A DO41                                                                                                                                       | <a href="#">Digikey</a> | 25 | \$0.16 | \$4.05 |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | RXEF050HF-<br>ND                               | RXEF050 PTC<br>RESET FUSE 72V<br>500MA RADIAL                                                                                                                       | <a href="#">Digikey</a> | 10 | \$0.35 | \$3.50 |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | 296-TPS25961<br>DRVRCT-ND                      | TPS25961DRVR<br>TPS25961<br>2.7-V–19-V, 106-mΩ,<br>eFuse With<br>Adjustable Current<br>Limit and<br>Short-Circuit<br>Protection                                     | <a href="#">Digikey</a> | 10 | \$0.51 | \$5.10 |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | BOJACK:<br>786118004542<br>ASIN:<br>B085RDTCCV | BOJACK 10 Values<br>300 Pcs Ceramic<br>Capacitor 0.1 0.15<br>0.22 0.33 0.47 0.68 1<br>2.2 4.7 10 uF<br>Multilayer Monolithic<br>Ceramic Capacitor<br>Assortment Kit | <a href="#">Amazon</a>  | 1  | \$9.99 | \$9.99 |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | ASIN:<br>B0C1VBXCQ<br>M                        | 24 Values Electrolytic<br>Capacitor Assortment<br>Kit from 0.1uF to<br>1000uF 10V 16V 25V<br>50V in-line<br>Aluminum Capacitors<br>Set                              | <a href="#">Amazon</a>  | 1  | \$9.99 | \$9.99 |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | ASIN:                                          | MECCANIXITY<br>10Pcs 2 Pin PCB<br>Mount Screw<br>Terminal Block<br>Connector 5.08mm<br>0.2 Inch Pitch PCB<br>Wiring Terminals                                       | <a href="#">Amazon</a>  | 1  | \$9.99 | \$9.99 |

|                                                |                                                           |                                                                                                                                                                                                                        |                        |   |         |             |
|------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---|---------|-------------|
|                                                |                                                           | 300V 10A for<br>22-12AWG Cable,<br>Green                                                                                                                                                                               |                        |   |         |             |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | ASIN:<br>B07TP2YYQB                                       | 320PCS M3 Male<br>Female Nylon Hex<br>Spacer Standoff<br>Screw Nut Threaded<br>Pillar PCB<br>Motherboard Assorted<br>Assortment Kit<br>(Black)                                                                         | <a href="#">Amazon</a> | 1 | \$9.99  | \$9.99      |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | ASIN:<br>B0746HG158                                       | Electrical Wire 22<br>AWG 22 Gauge<br>Silicone Wire Hook<br>Up Wire Cable 20<br>Feet [10 ft Black and<br>10 ft Red] - Soft and<br>Flexible 60 Strands<br>0.08mm of Tinned<br>Copper Wire High<br>Temperature Resistant | <a href="#">Amazon</a> | 1 | \$5.18  | \$5.18      |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | ZSDZFYLLK:<br>ZSDZFYLLK-<br>2P-2.0<br>ASIN:<br>B0D7Q9HJLQ | 20 Sets Mini Micro<br>JST-PH 2.0mm 2 Pin<br>Male Connector with<br>10cm Silicone Wire<br>Cables and Female<br>Socket JST PH 2.0<br>Connector Kit                                                                       | <a href="#">Amazon</a> | 1 | \$6.99  | \$6.99      |
| <i>University<br/>of Houston<br/>&amp; FDA</i> | PCEONAMP:<br>XY11202J<br>ASIN:<br>B0DWJN8GZ<br>3          | SCX24 Battery<br>1150mAh 2S 7.4V<br>Lipo Battery<br>20C(Burst 40C) with<br>PH2.0 Plug for<br>SCX24 RC Models<br>with USB Charger                                                                                       | <a href="#">Amazon</a> | 1 | \$23.99 | \$23.9<br>9 |

## **H. Design Deliverables**

The following deliverables were completed:

- System architecture and block diagrams
- Subsystem schematics
- Engineering specifications and constraints
- Trade studies and component selection
- Preliminary bill of materials
- Procurement plan

## **IV. Development Phase (In Progress)**

The next phase involves translating the design into a functional prototype. This includes hardware assembly, subsystem integration, and firmware implementation. Emphasis will be placed on validating signal integrity and RF suppression mechanisms. Currently the team is in the process of ordering parts, and have a bill of materials already. However, the display software is already completed and functional.

## **V. Implementation Phase (Planned)**

The system will be deployed in a simulated MRI environment using EMF generators and AIMD samples. The implementation will focus on usability, repeatability, and safe operation.

## **VI. Evaluation Phase (Planned)**

System performance will be evaluated through:

- Controlled RF signal testing
- Measurement accuracy verification
- Filter and ADC validation
- Data integrity checks

Success will be determined by the system's ability to produce accurate, stable, and repeatable measurements under RF exposure conditions.

## **VII. Conclusion**

This report documents the successful completion of the Analysis Design, and Development phases of the ADDIE framework for an RF-induced voltage detector. The proposed system addresses key challenges related to RF interference, measurement accuracy, and usability. With a finalized design and procurement plan, the project is prepared to transition through the

Development phase, and into the Implementation phase, where the system will be implemented and validated.

### **References**

[1] ISO 10974, *Assessment of the safety of magnetic resonance imaging for patients with an active implantable medical device*, International Organization for Standardization.